Overtime cost is rising on paper, not on the shop floor

Indus Weave Textiles - HR data pipeline, 1 Apr to 30 Sep 2026

+16%

paid overtime hours Apr to Sep
(5,526 to 6,419)
hours actually worked: -2%

PKR 4.8M

paid to 11 people with
no attendance in the pay month

60 min

SITE punch clock error from 1 July
(2,673 of 2,801 SITE working-day
overtime requests reconcile once fixed)

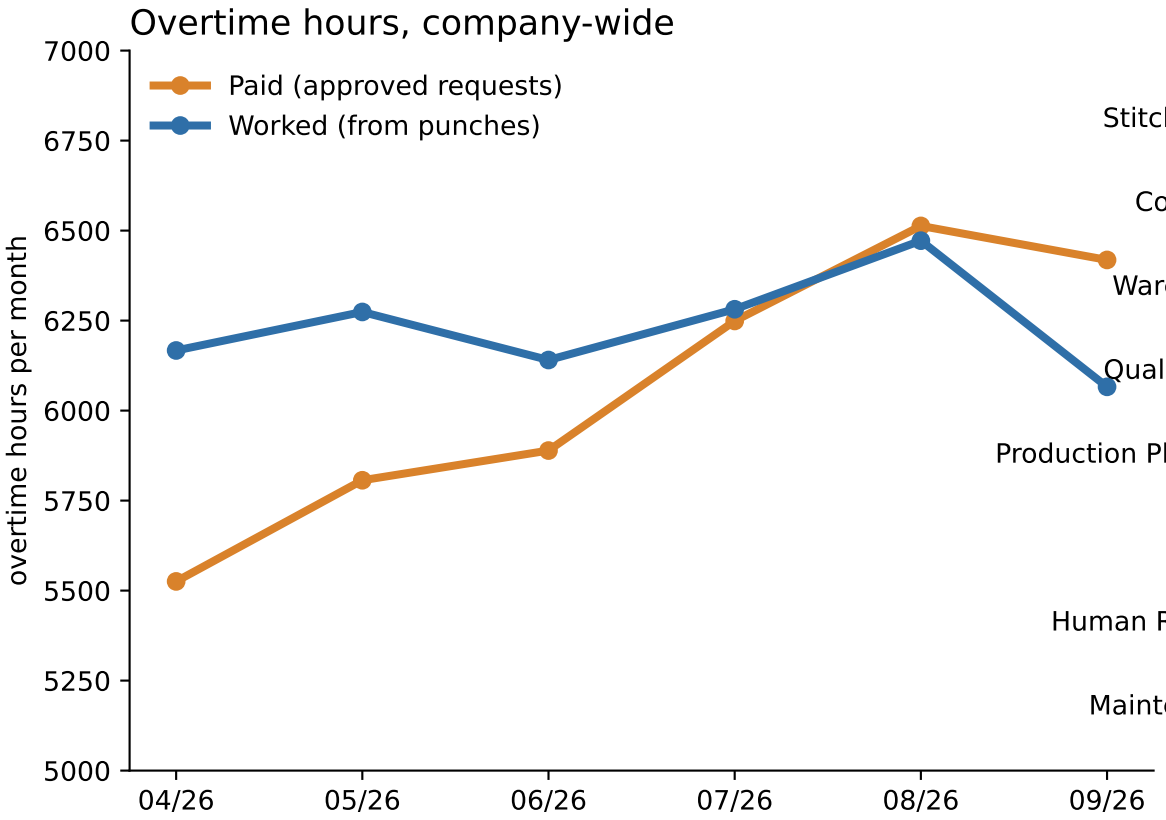
The overtime gap sits in one department, Faisalabad Stitching, under one approver.
Payroll pays from approved requests, not from punches, so nobody checks the hours.

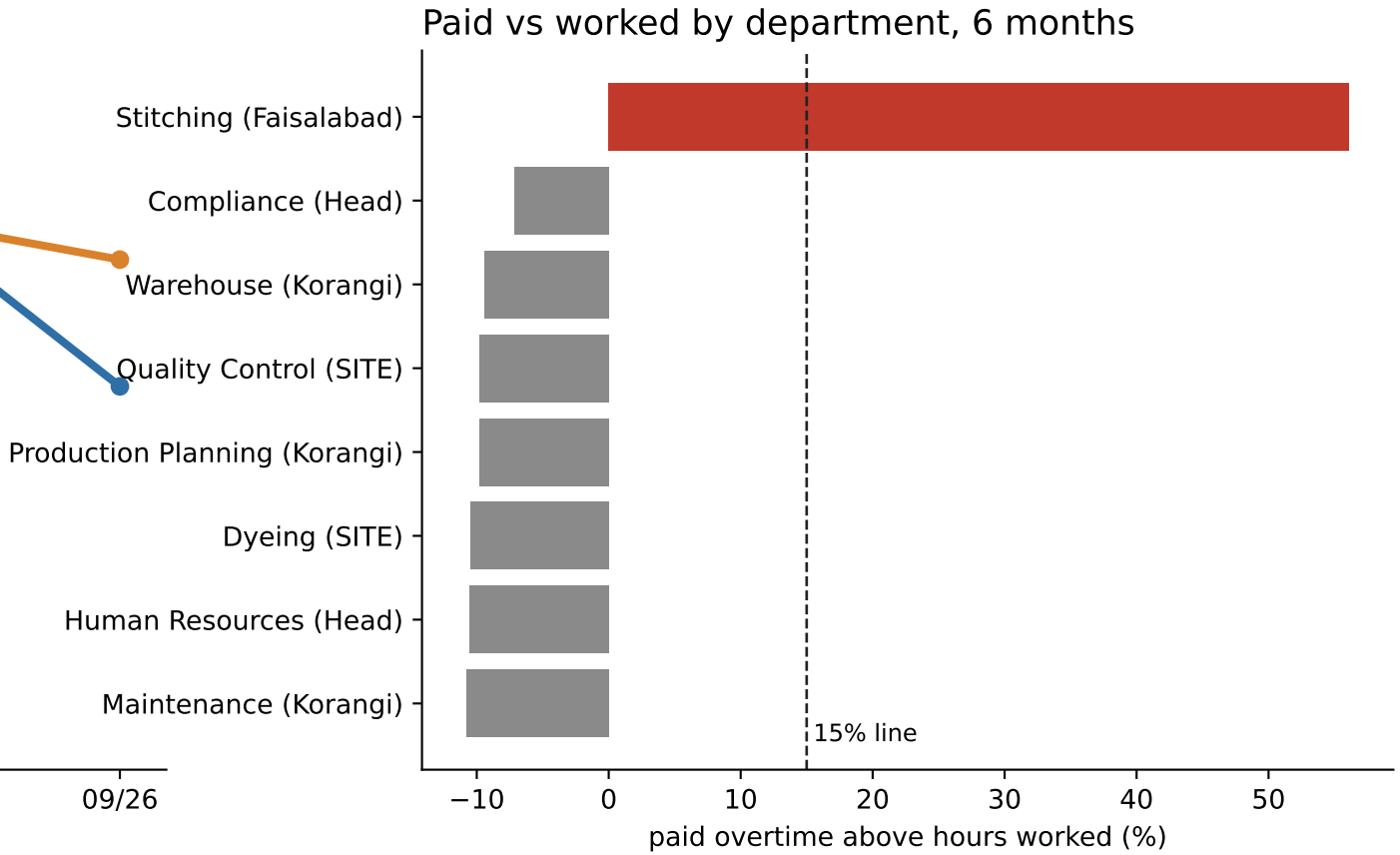
All numbers come from a rebuilt warehouse: 12 raw files -> clean model -> 10 SQL answers,
174/174 data quality checks passing.

Source: indus_weave_hcm_data, pipeline in this repository

Finding 1: we pay for overtime that punches do not support

Paid hours up 16%, worked hours -2%. The gap sits in one department.





Faisalabad Stitching: 7,659 h paid vs 4,906 h worked (+56%), over the 15% line in 5 of 6 months. Approver E10382 approved all 3,174 Stitching requests; 2,048 of them (158 employees) exceed the hours worked, 3,357 h or PKR 1.7M in total. No other department has any.

Evidence: analytics/q05, q06; presentation/evidence/f1_*.sql

Finding 2: PKR 4.8M went to people with no attendance

Paid in a month in which they did not attend a single day

PKR 1.9M

5 employees paid in full for 21 pay-months after their last working day

PKR 2.9M

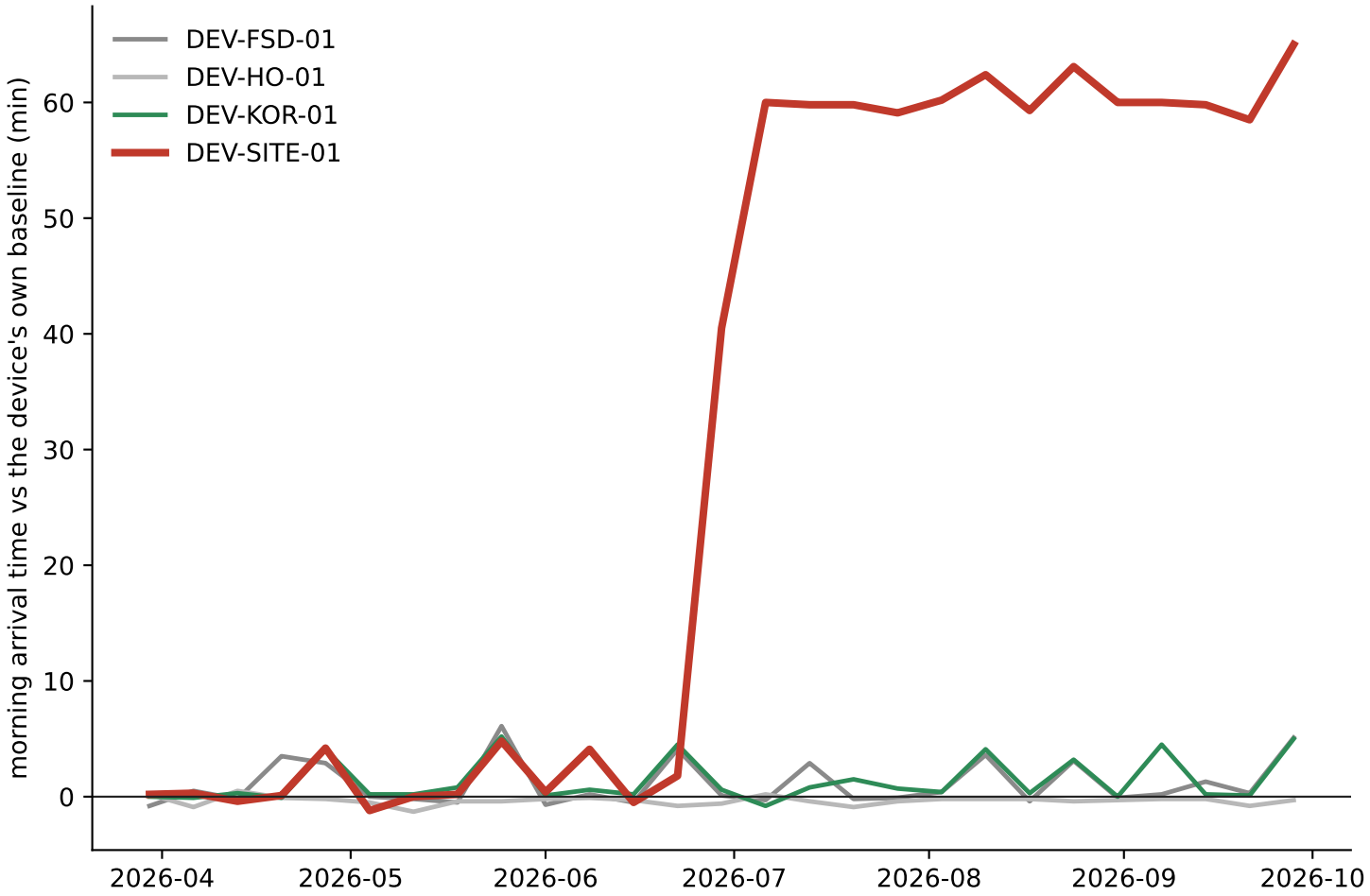
6 active employees, 36 pay-months, no biometric device and no punch in six months

Employee	Name	Department	Last day worked	Months paid after	Gross paid after (PKR)
E10471	Umair Baloch	Human Resources	2026-04-23	5	660,000
E11222	Naveed Ansari	Dyeing	2026-05-16	4	418,100
E10332	Amna Awan	Warehouse	2026-05-14	4	322,000
E10989	Tariq Jatoi	Packing	2026-05-18	4	294,100
E10295	Sohail Hussain	Finishing	2026-05-11	4	253,800

Evidence: analytics/q07, q08; presentation/evidence/f2_pay_without_attendance.sql

Finding 3: one punch clock ran an hour fast

DEV-SITE-01 was exactly 60 minutes ahead from 1 July, so SITE numbers could not be trusted; the other devices stayed flat



95.4%

of SITE working-day overtime requests (1 Jul on; 2,801) reconcile after the fix

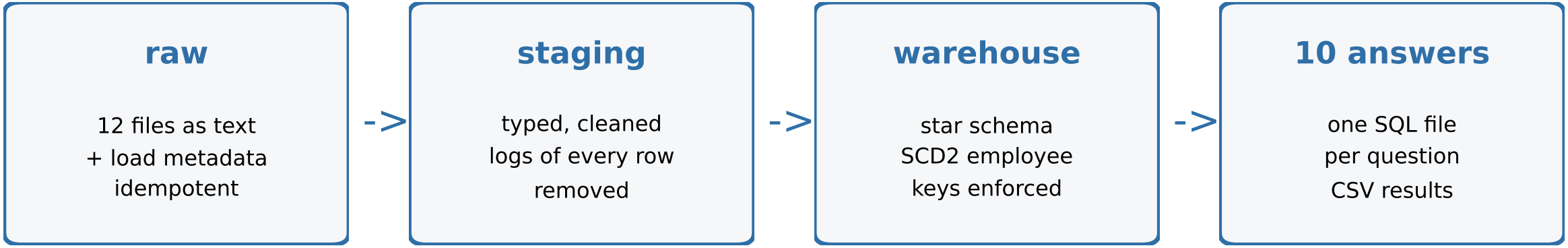
0%

reconcile on the raw, uncorrected SITE punch file

Found by comparing each device's morning arrival edge with its own baseline; 45,803 punches corrected. Evidence: presentation/evidence/f3_clock_drift.sql

How we know the numbers are right

Every raw row is accounted for, and the checks run on every load



- 174 of 174 automated checks pass (keys, foreign keys, accepted values, business sanity, SCD2 ranges)
- raw rows = staging + rejected + merged, for every table (345,280 punches = 322,143 clean + 6,962 unmapped + 16,175 duplicates)
- every one of the 322,143 clean punches lands on exactly one attendance day
- payroll overtime hours = approved overtime hours in all 6,572 employee-months: payroll follows requests, not punches
- the whole warehouse rebuilds from the raw files with one command: python run_pipeline.py

quality/dq_checks.py writes every result to dq.dq_results

What we recommend

Three actions, in order of money at stake

1

Pay overtime only for hours the punches support

Stop payment on requests whose approved hours exceed hours worked. Review the PKR 1.7M already paid in Faisalabad Stitching and the approvals of E10382. Make approval depend on a punch check.

2

Recover and stop the PKR 4.8M paid without attendance

Remove the 5 leavers from payroll and recover PKR 1.9M. Verify that the 6 active employees without a device exist and work, then enrol them.

3

Make the data trustworthy every month

Run this pipeline on each export. Alert when a device drifts, when a leaver is paid, or when paid overtime exceeds worked overtime by more than 15%.

Details and evidence queries: presentation/cfo_findings.md